Review draft - 27 October 2026. The original pooled-analysis claims are withdrawn pending verification. Informal evidence categories are not formal GRADE assessments. Remaining study-level statements and citations require audit before submission or clinical interpretation. The PDF is rebuilt from this revised Markdown source and remains a review candidate.

Perioperative Analgesic Strategy in Total Knee Arthroplasty: A Scoping Review Across Anesthesia Modality, Regional Techniques, Adjuvant Pharmacology, Risk Stratification, and Outcome Measurement

With evidence gap analysis for robot-assisted total knee arthroplasty

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Abstract

Background: Perioperative analgesic planning in total knee arthroplasty (TKA) spans five sequential clinical decision domains - anesthesia modality, regional analgesic technique, adjuvant pharmacology, patient risk phenotype, and outcome measurement - that have been studied largely in isolation. No prior review has mapped the scope of evidence across all five domains simultaneously or systematically examined the implications of that evidence for robot-assisted TKA (rTKA), a platform for which direct perioperative analgesic evidence is absent.

Objective: To map and synthesize the scope, nature, and quality of evidence for perioperative analgesic strategies across five decision domains in adults undergoing primary TKA, and to characterize evidence gaps relevant to robot-assisted TKA.

Eligibility criteria: English-language systematic reviews, meta-analyses, RCTs, prospective cohort studies, and major guideline documents addressing perioperative analgesic strategy in adults undergoing primary TKA. Case reports, editorials, letters, and conference abstracts without full data were excluded.

Information sources: PubMed/MEDLINE searched through March 21, 2026, supplemented by targeted retrieval of guideline documents from ERAS Society, AAHKS/AAOS/ASRA, ASA, and ASA Task Force publications, and backward reference tracking from foundational systematic reviews and meta-analyses.

Selection and charting: Single-reviewer title/abstract screening and full-text assessment. Data were charted by decision domain, study design, population, intervention, comparator, and outcome. Evidence quality was assessed using a informal author-assigned evidence categories (not a formal GRADE assessment).

Results (revision status): The draft maps five domains but its source count, extraction tables, and original pooled estimates require independent verification. The previous inference of non-inferiority from nonsignificant differences has been withdrawn. No new pooled result is asserted in this revision.

Conclusion: The perioperative analgesic literature in TKA contains substantial but fragmented evidence across five clinical decision domains. Risk stratification and outcome measurement are the most evidence-deficient domains relative to their clinical importance. Direct rTKA evidence was not established by the documented search. A risk-stratified, measurement-aware perioperative framework is warranted and should be evaluated prospectively in rTKA-specific controlled trials.

Keywords: total knee arthroplasty; scoping review; perioperative analgesia; adductor canal block; IPACK; opioid-free anesthesia; pain catastrophizing; patient-reported outcomes; MCID; robot-assisted arthroplasty; PRISMA-ScR

1. Introduction

Total knee arthroplasty is among the most frequently performed elective surgical procedures in the United States, with volumes projected to exceed four million cases annually by 2040. Clinical outcomes at the population level are well-established: TKA reliably reduces joint pain and improves function in appropriately selected patients with end-stage knee osteoarthritis. Yet patient-reported dissatisfaction after TKA persists in 15-25% of recipients, and chronic postoperative pain - defined as pain persisting beyond three months at the operative site - affects up to 20% of TKA patients across prospective cohort data. These rates have not improved consistently despite decades of surgical and anesthetic refinement.

The perioperative analgesic literature in TKA has addressed this problem in isolated fragments. Anesthesia modality trials compare spinal to general anesthesia on PONV, blood loss, and 30-day morbidity. Regional analgesia trials compare ACB to FNB on NRS scores and TUG performance. Adjuvant pharmacology trials compare opioid-free anesthesia protocols to standard care on 24-hour MME consumption. Risk stratification studies identify psychological and pharmacological predictors of poor outcomes. Outcomes measurement researchers evaluate instrument responsiveness and MCID calibration. No prior review has systematically mapped the scope of evidence across all five domains simultaneously or characterized the evidence gaps relevant to robot-assisted TKA, a platform that now represents a significant and growing proportion of TKA volume in many centers.

This scoping review maps evidence across five perioperative analgesic decision domains in adults undergoing primary TKA, synthesizes that evidence by domain, characterizes evidence quality and gaps, and reports evidence gap implications for robot-assisted TKA where direct evidence is absent. The rTKA application layer is explicitly designated as evidence-gap analysis and inference rather than established evidence.

2. Methods

This scoping review was conducted in accordance with JBI scoping review methodology and is reported according to the PRISMA extension for Scoping Reviews (PRISMA-ScR). A PRISMA-ScR checklist is provided as Supplementary File 1.

2.1 Review Design

This is a scoping review following JBI guidance for scoping reviews. The purpose is to map and synthesize the nature and scope of evidence for perioperative analgesic strategies across five clinical decision domains in TKA, not to conduct a pooled meta-analysis or provide a systematic assessment of all available studies on any single intervention. In accordance with JBI scoping review methodology, formal risk-of-bias scoring was not performed because the objective was evidence mapping and structured synthesis across decision domains rather than pooled effect estimation. A informal author-assigned evidence categories (not a formal GRADE assessment) was applied to the mapped evidence to characterize relative evidence quality across included source types; this overlay is supplementary to the scoping framework, not a replacement for it.

2.2 Review Question

Using the Population-Concept-Context (PCC) framework:

- **Population:** Adults (age ≥ 18 years) undergoing primary elective total knee arthroplasty for knee osteoarthritis.

- **Concept:** Perioperative analgesic strategies across five decision domains: (1) anesthesia modality (spinal versus general anesthesia), (2) regional analgesic technique (adductor canal block, IPACK, periarticular infiltration analgesia, femoral nerve block), (3) intraoperative adjuvant pharmacology (ketamine, dexmedetomidine, IV lidocaine, magnesium, NSAIDs/COX-2 inhibitors, dexamethasone, IV acetaminophen), (4) patient risk stratification (preoperative opioid use, pain catastrophizing, central sensitization, depression, anxiety, sleep disturbance), and (5) outcome measurement (KOOS, WOMAC, PROMIS, NRS, TUG, MCID calibration, PGIC).

- **Context:** Perioperative and postoperative analgesic management in primary TKA performed in hospital or ambulatory surgery settings, including evidence gap analysis for robot-assisted TKA where direct evidence is limited or absent.

The overarching review question is: *What is the scope, nature, and quality of evidence for perioperative analgesic strategies across the five identified clinical decision domains in adults undergoing primary TKA, and what are the evidence gaps and portability implications for robot-assisted TKA?*

2.3 Information Sources

PubMed/MEDLINE was searched as the primary bibliographic database. Searches were conducted through March 21, 2026, with no lower date limit applied. Searches were restricted to English-language publications. In addition to database searching, targeted retrieval was conducted for guideline documents and consensus statements from the following sources: ERAS Society, American Association of Hip and Knee Surgeons (AAHKS), American Academy of Orthopaedic Surgeons (AAOS), American Society of Regional Anesthesia and Pain Medicine (ASRA), American Society of Anesthesiologists (ASA), and ASA Task Force publications. Backward reference tracking from identified systematic reviews, meta-analyses, and guideline-grade documents was performed to capture foundational studies not identified in initial database searches. No additional databases (Embase, Cochrane Library, Scopus, Web of Science) were searched; this is acknowledged as a limitation (see Section 9).

2.4 Search Strategy

Searches were conducted in PubMed/MEDLINE using the following structured queries, executed through March 21, 2026. Search terms were iteratively refined across domains.

Domain 1 - Anesthesia modality: ("total knee arthroplasty"[MeSH] OR "total knee replacement"[tiab]) AND ("spinal anesthesia"[MeSH] OR "general anesthesia"[MeSH] OR "neuraxial"[tiab]) AND ("postoperative

complications"[MeSH] OR "pain, postoperative"[MeSH] OR "nausea and vomiting, postoperative"[MeSH])

Domain 2 - Regional analgesia: ("total knee arthroplasty"[MeSH] OR "total knee replacement"[tiab]) AND ("adductor canal block"[tiab] OR "femoral nerve block"[MeSH] OR "nerve block"[MeSH] OR "IPACK"[tiab] OR "periarticular infiltration"[tiab])

Domain 3 - Adjuvant pharmacology: ("total knee arthroplasty"[MeSH] OR "arthroplasty, replacement, knee"[MeSH]) AND ("ketamine"[MeSH] OR "dexmedetomidine"[MeSH] OR "lidocaine"[MeSH] OR "magnesium"[MeSH] OR "anti-inflammatory agents, non-steroidal"[MeSH] OR "dexamethasone"[MeSH] OR "acetaminophen"[MeSH]) AND ("pain management"[MeSH] OR "analgesics, opioid"[MeSH])

Domain 4 - Risk stratification: ("total knee arthroplasty"[MeSH] OR "total knee replacement"[tiab]) AND ("catastrophization"[MeSH] OR "pain catastrophizing"[tiab] OR "central sensitization"[tiab] OR "opioid"[tiab] OR "depression"[MeSH] OR "anxiety"[MeSH] OR "sleep"[MeSH]) AND ("treatment outcome"[MeSH] OR "prognosis"[MeSH])

Domain 5 - Outcome measurement: ("total knee arthroplasty"[MeSH] OR "total knee replacement"[tiab]) AND ("patient reported outcome measures"[MeSH] OR "KOOS"[tiab] OR "WOMAC"[tiab] OR "PROMIS"[tiab] OR "minimal clinically important difference"[tiab] OR "MCID"[tiab])

Guideline document retrieval used organizational website searches and PubMed queries with organization names as author fields or publication source terms.

2.5 Eligibility Criteria

Inclusion criteria:

- Study design: systematic reviews, meta-analyses, randomized controlled trials, prospective cohort studies, and guideline-grade consensus documents
- Population: adults (age ≥ 18 years) undergoing primary elective TKA
- Language: English only
- Date: no lower limit; searches conducted through March 21, 2026
- Scope: perioperative analgesic strategy in at least one of the five defined domains
- Indexing: all included studies are indexed in PubMed with available PMIDs

Exclusion criteria:

- Case reports, case series (fewer than 20 patients), editorials, letters, and opinion pieces
- Conference abstracts without full-text published data
- Studies addressing revision TKA, unicompartmental arthroplasty, or total hip arthroplasty as the primary population unless reporting TKA subgroup data analyzed separately
- Studies addressing pediatric populations
- Non-English language publications

2.6 Selection Process

Title and abstract screening was performed by a single reviewer (CBG) against the eligibility criteria. Full-text assessment was performed for all sources passing title/abstract screening. Single-reviewer screening is a methodological limitation of this review and is reported transparently in the Limitations section (9.5). No duplicate independent screening was performed; no conflict resolution process was therefore required. The PRISMA flow diagram is presented as Supplementary File 2.

2.7 Data Charting

A standardized data charting form was applied to each included source. The following variables were extracted:

- Study design (RCT, systematic review/meta-analysis, prospective cohort, guideline document)
- Population (TKA-specific or mixed arthroplasty with TKA subgroup)
- Intervention and comparator
- Primary outcome domain (pain, functional recovery, opioid consumption, adverse events)
- Risk phenotype variables addressed (if applicable)
- Outcome measurement instruments used
- Relevance to robot-assisted TKA portability
- Evidence tier (Tier 1: RCTs, systematic reviews, meta-analyses, guideline-grade documents; Tier 2: observational, contextual, or corroborating evidence)

2.8 Synthesis

Included sources were synthesized by perioperative decision domain. Within each domain, evidence was organized by study design hierarchy, with systematic reviews and meta-analyses as the primary synthesis source and individual RCTs cited where foundational evidence has not yet been superseded by higher-level synthesis. The earlier original pooling is withdrawn pending verification; this revision reports no new quantitative synthesis.

3. Results: Anesthesia Modality - Spinal Versus General Anesthesia

3.1 Evidence Scope

Included sources for this domain: 10 studies and guideline documents (Table 1). Study designs: 1 RCT (REGAIN trial, hip fracture population), 2 systematic reviews/meta-analyses (Johnson et al. 2016; Zywiel et al. 2014), 1 Cochrane review (Hu et al. 2009), 2 large observational database studies (Memtsoudis et al. 2013; Stundner et al. 2012), 2 guideline documents (Wainwright/ERAS Society 2020; Horlocker/ASRA 2018), and 2 clinical practice standards (Gonvers et al. 2021; ASA neuraxial opioids 2016). The absence of an adequately powered RCT comparing spinal to general anesthesia specifically in elective TKA with anesthesia modality as the primary endpoint is the dominant evidence gap for this domain.

Table 1. Evidence map - Anesthesia modality domain

Source: Wainwright et al. 2020[3]
Design: Guideline (ERAS Society)
N: -

Primary endpoint: Modality recommendation
Informal category: B
Key finding: Neuraxial preferred where not contraindicated

Source: Johnson et al. 2016[5]
Design: Systematic review/MA
N: Mixed THA/TKA
Primary endpoint: PONV, functional outcomes
Informal category: B
Key finding: No statistically significant PONV difference in pooled analysis

Source: Gonvers et al. 2021[6]
Design: Systematic review/MA
N: Hip and knee arthroplasty
Primary endpoint: 24-hr opioid consumption
Informal category: B
Key finding: ITM reduces postoperative opioid use; monitoring burden documented

Source: Hu et al. 2009[8]
Design: Cochrane review
N: Hip and knee arthroplasty
Primary endpoint: Blood loss, transfusion
Informal category: B (era-dependent)
Key finding: Regional anesthesia reduced blood loss; pre-TXA era data

Source: Fillingham et al. 2018[9]
Design: Guideline (AAHKS)
N: -
Primary endpoint: Transfusion management
Informal category: A
Key finding: TXA standard of care; attenuates neuraxial blood loss advantage

Source: Memtsoudis et al. 2013[10]
Design: Observational database
N: 382,236 arthroplasty
Primary endpoint: Morbidity, mortality
Informal category: C
Key finding: Directional neuraxial advantage; substantial confounding

Source: Neuman et al. 2021[11]
Design: RCT (hip fracture)
N: 1,600
Primary endpoint: Ambulation, complications
Informal category: B (indirect)
Key finding: No significant difference on major outcomes by modality

Source: Zywiel et al. 2014[12]
Design: Systematic review
N: Elective joint arthroplasty
Primary endpoint: POCD
Informal category: C
Key finding: Inconsistent evidence; no neuraxial advantage established

Source: Horlocker et al. 2018[13]
Design: Guideline (ASRA 4th ed.)
N: -
Primary endpoint: Neuraxial safety
Informal category: A
Key finding: Anticoagulation contraindication framework

Source: Stundner et al. 2012[60]
Design: Observational
N: Bilateral TKA
Primary endpoint: Perioperative outcomes
Informal category: C
Key finding: Directional neuraxial preference; confounding risk elevated

3.2 Principal Findings

The ERAS Society guidelines[3] recommend neuraxial anesthesia as the preferred modality where not contraindicated. The evidence underlying this recommendation stratifies into three tiers of quality.

The mechanistically favoured tier - PONV reduction and early opioid exposure attenuation - carries informal category B support. The proposed mechanism (elimination of volatile anesthetic and intraoperative opioid as Apfel risk factors[4]) is robust and platform-independent. The best available comparative synthesis (Johnson et al.[5]) found no statistically significant difference in pooled PONV incidence, indicating that the mechanism-based advantage is attenuated in antiemetic-treated contemporary pathways. Intrathecal morphine (ITM) reduces 24-hour opioid consumption (Gonvers et al.[6], informal category B) but imposes a monitoring burden[7] that conflicts with short-stay ERAS protocols.

The qualified tier - blood loss and transfusion reduction - carries informal category B evidence that is era-dependent and probably attenuated in contemporary TXA-standardized practice[9].

The weak-evidence tier - thromboembolic risk, major morbidity, mortality, and POCD - does not carry controlled evidence of the quality required to support confident claims. Observational data (Memtsoudis et al.[10]) are directionally consistent but structurally confounded; the REGAIN trial[11] found no significant major outcome differences by modality under controlled conditions.

3.3 Key Evidence Gaps

No RCT has evaluated spinal versus general anesthesia with anesthesia modality as the primary endpoint in elective conventional TKA or rTKA. Effect sizes for blood loss advantage are pre-TXA era data not representative of contemporary practice. The ITM monitoring-versus-discharge-readiness tradeoff in short-stay arthroplasty has not been evaluated in a controlled trial.

3.4 Evidence Gap - Implications for Robot-Assisted TKA

No evidence addresses anesthesia modality in rTKA. The following implications are EVIDENCE GAP - ANALYTICALLY INFERRED. PONV reduction and ITM opioid-sparing are mechanism-based and platform-independent; these advantages are likely preserved in rTKA. Blood loss reduction may be preserved in tourniquet-free rTKA through sympathectomy-mediated hemostasis, but the absolute clinical magnitude is probably further attenuated by rTKA's independently lower blood loss profile. ITM monitoring requirements create a practical tension with short-stay rTKA discharge protocols that has not been evaluated.

The following section presents the mapped evidence for regional analgesic technique selection within the established anesthesia platform.

4. Regional analgesia: corrected selected-source interpretation

The earlier regional evidence table contained bibliographic mismatches and unsupported certainty claims. It is replaced with this narrower interpretation; no new systematic search or pooling is claimed.

4.1 Adductor canal versus femoral nerve block

The relevant Cochrane review is Schnabel et al. (2019), reference 17. It emphasizes outcome-specific uncertainty. Its findings do not establish universal analgesic non-inferiority, equivalence, or a reduction in falls. A lack of a detected difference is not itself an equivalence or non-inferiority result. Motor-strength findings from individual studies should be distinguished from pain and longer-term functional outcomes.

4.2 iPACK and periarticular infiltration

Hussain et al. (2021), reference 20, found that adding iPACK to ACB did not improve analgesic outcomes when periarticular local infiltration analgesia was already used. Without that infiltration, pain and functional outcomes improved in the analyzed trials, but opioid consumption did not. The earlier draft reversed this background-treatment qualification and overstated opioid reduction; that interpretation is withdrawn.

4.3 Dexamethasone and applicability

Reference 23 is Albrecht et al., published in 2015, not the cited 2014 volume/pages. Its prolonged-analgesia findings were highly heterogeneous and concern multiple peripheral blocks; a fixed block-duration gain should not be generalized to ACB or robot-assisted TKA.

4.4 Limits

These selected checks do not verify every trial previously listed. No robot-specific efficacy, optimal injection regimen, preferred technique, or certainty grade is established here. A complete regional review requires a new source-level audit before reinstating the omitted evidence table.

5. Results: Adjuvant Pharmacology and Opioid-Minimizing Pathways

5.1 Evidence Scope

Included sources for this domain: 16 studies and guideline documents (Table 3). Study designs: 3 guideline-grade systematic reviews (Hannon et al. 2023 ketamine; Fillingham et al. 2020 NSAIDs; Waldron et al. 2013 dexamethasone), 6 systematic reviews and meta-analyses across agents, 2 TJA-specific meta-analyses (Xu et al. acetaminophen), 1 RCT (PRECISION cardiovascular safety), and 4 mixed-surgical-population systematic reviews.

Table 3. Evidence map - Adjuvant pharmacology domain

Agent: NSAIDs/COX-2
Primary source(s): Fillingham et al. 2020[31]
Design: Guideline SR (AAHKS/AAOS/ASRA)
Informal category: A
Key finding: "Strong evidence" for opioid-sparing and pain reduction; PRECISION[32] establishes COX-2 cardiovascular safety

Agent: Ketamine
Primary source(s): Hannon et al. 2023[24]; Schwenk et al. 2021[26]
Design: Guideline SR; clinical review
Informal category: B
Key finding: Reduces 24-hr MME; strongest effect in opioid-tolerant patients; ceiling effect in opioid-naive neuraxial pathway

Agent: Dexamethasone
Primary source(s): Waldron et al. 2013[33]
Design: Meta-analysis
Informal category: B
Key finding: Analgesic and antiemetic effects established; mechanisms distinct

Agent: Dexmedetomidine
Primary source(s): Yang et al. 2020[27]; Blaudszun et al. 2012[28]
Design: Meta-analyses (TKA/THA; mixed surgical)
Informal category: B
Key finding: Opioid-sparing in TKA/THA; bradycardia in TKA subgroup limits use

Agent: IV acetaminophen
Primary source(s): Xu et al. 2017[34]; Xu et al. 2018[35]
Design: Meta-analyses
Informal category: B
Key finding: Opioid-sparing established; no IV-over-oral advantage in elective TJA

Agent: IV lidocaine
Primary source(s): Vigneault et al. 2011[29]
Design: Systematic review
Informal category: C
Key finding: Opioid-sparing in mixed surgical populations; no TKA-specific controlled trial with neuraxial+block comparator

Agent: Magnesium
Primary source(s): Albrecht et al. 2013[30]
Design: Meta-analysis
Informal category: C
Key finding: Statistically significant but clinically modest effect; no arthroplasty-specific controlled evidence

5.2 Principal Findings

The evidence gradient across adjuvant agents is pronounced. NSAIDs and COX-2 inhibitors carry the strongest evidence (informal category A) supported by a guideline-grade systematic review and an adequately powered cardiovascular safety trial. Ketamine (informal category B), dexamethasone (informal category B), dexmedetomidine (informal category B), and IV acetaminophen (informal category B) are each supported by arthroplasty-specific meta-analytic evidence. IV lidocaine and magnesium (both informal category C) lack TKA-specific controlled evidence against a contemporary neuraxial-plus-block comparator.

A critical interpretive issue applies across this domain: the ceiling effect problem. The benefit of any individual adjuvant agent is highest when background analgesic coverage is lowest, and is attenuated when neuraxial anesthesia and regional block techniques are already providing near-complete coverage of the relevant pain generators. Effect sizes from OFA trials conducted without regional block coverage are not directly portable to contemporary neuraxial-plus-ACB pathways. This interpretive constraint is not consistently acknowledged in the primary OFA trial literature and constitutes a systematic bias in the available evidence base for this domain.

A second interpretive issue applies to bundle-level OFA evidence: when OFA experimental arms include neuraxial anesthesia, ACB, NSAIDs, dexamethasone, and ketamine while comparator arms include fewer components, observed bundle-level differences cannot be attributed to the opioid elimination strategy per se. Agent-specific evidence is the appropriate analytic unit.

5.3 Key Evidence Gaps

No adequately powered TKA-specific RCT has evaluated IV lidocaine against a contemporary neuraxial-plus-regional-block comparator. Magnesium sulfate lacks arthroplasty-specific controlled evidence beyond historical single-center data. Ketamine's ceiling effect in opioid-naive patients on contemporary neuraxial-plus-block pathways has not been formally evaluated in a stratified RCT. Bundle-level OFA evidence does not permit agent-level attribution.

5.4 Evidence Gap - Implications for Robot-Assisted TKA

Pharmacological mechanisms are platform-independent. Evidence grades assigned above port to rTKA without modification. Agents whose effect sizes are primarily driven by rescue of inadequate analgesia in high-pain operative contexts may show further attenuation in rTKA if rTKA's baseline pain burden is independently lower; this prediction applies most directly to IV lidocaine (which lacks even conventional TKA evidence) and to agents with ceiling effects in neuraxial-plus-block pathways.

The effect sizes reported in Sections 3-5 were derived from populations that include but do not separately analyze patients with preoperative opioid tolerance, central sensitization, pain catastrophizing, and psychological comorbidity. The following section presents the mapped evidence for preoperative risk stratification in those subgroups.

6. Results: Risk-Stratified Perioperative Planning

6.1 Evidence Scope

Included sources for this domain: 14 studies (Table 4). Study designs: 3 systematic reviews (Burns et al. 2015; Vissers et al. 2012; Malahias et al. 2019), 5 prospective cohort studies (Smith et al. 2017; Ross et al. 2021; Forsythe et al. 2008; Arendt-Nielsen et al. 2010; Wylde et al. 2011), 3 prospective observational studies (Lingard et al. 2004; Brander et al. 2003; Cremeans-Smith et al. 2006), 2 longitudinal cohort studies (Luo et al. 2019; Zywiel et al. 2011), and 1 landmark framework paper (Kehlet et al. 2006).

Table 4. Evidence map - Risk stratification domain

Risk domain: Preoperative opioid use - outcomes
Source(s): Smith et al. 2017[36]
Design: Prospective cohort
Informal category: A
Key finding: Preoperative opioid users: significantly worse 6-month pain and function after TKA

Risk domain: Preoperative opioid use - dose-response
Source(s): Ross et al. 2021[37]
Design: Retrospective cohort
Informal category: A
Key finding: Dose-dependent worsening: even low daily exposures associated with worse outcomes

Risk domain: Preoperative opioid tapering
Source(s): Zywiel et al. 2011[38]
Design: Observational cohort
Informal category: B
Key finding: Preoperative opioid cessation associated with improved functional outcomes

Risk domain: Pain catastrophizing
Source(s): Forsythe et al. 2008[39]
Design: Prospective cohort
Informal category: B
Key finding: PCS independently predicts 6-month postoperative pain after TKA

Risk domain: Psychological predictors (systematic)
Source(s): Burns et al. 2015[40]
Design: Systematic review
Informal category: B
Key finding: Catastrophizing is the most consistent psychological predictor of worse TKA pain outcomes

Risk domain: Central sensitization / CPM
Source(s): Arendt-Nielsen et al. 2010[41]
Design: Prospective cohort
Informal category: B
Key finding: Reduced CPM efficiency preoperatively predicts higher postoperative pain scores

Risk domain: Psychological factors (systematic)
Source(s): Vissers et al. 2012[42]
Design: Systematic review
Informal category: B
Key finding: Depression and anxiety independently predict worse TKA outcomes

Risk domain: Persistent postoperative pain
Source(s): Wylde et al. 2011[43]
Design: Prospective cohort
Informal category: B
Key finding: Preoperative widespread pain independently predicts chronic postoperative pain

Risk domain: Persistent postsurgical pain framework
Source(s): Kehlet et al. 2006[44]
Design: Review
Informal category: B
Key finding: Acute pain severity predicts chronic postsurgical pain; risk factors identified

Risk domain: Depression and TKA outcomes
Source(s): Lingard et al. 2004[45]
Design: Prospective cohort
Informal category: B
Key finding: Depression predicts lower KOOS at 12 months after TKA

Risk domain: Anxiety and TKA pain
Source(s): Brander et al. 2003[46]
Design: Prospective cohort
Informal category: B
Key finding: Preoperative anxiety predicts pain persistence at 2 years

Risk domain: Sleep disturbance and pain
Source(s): Cremeans-Smith et al. 2006[47]
Design: Prospective cohort
Informal category: B
Key finding: Sleep disruption mediates relationship between catastrophizing and postoperative pain

Risk domain: Sleep quality and TJA outcomes
Source(s): Luo et al. 2019[48]
Design: Prospective cohort
Informal category: B
Key finding: Preoperative sleep quality independently predicts postoperative pain, function, and LOS

Risk domain: Staged bilateral TKA
Source(s): Malahias et al. 2019[49]
Design: Systematic review
Informal category: B
Key finding: 5 of 7 studies report inferior outcomes on the second-operated knee in staged bilateral TKA

6.2 Principal Findings

The mapped evidence establishes five preoperative risk domains with independent predictive validity for worse TKA analgesic outcomes: preoperative opioid use (informal category A, dose-response established), pain catastrophizing (informal category B, most consistent predictor across systematic reviews), central sensitization (informal category B, CPM efficiency as preoperative marker), depression and anxiety (informal category B, multiple independent prospective cohorts), and sleep disturbance (informal category B, prospective cohort evidence).

The risk prediction evidence is substantially stronger than the evidence that specific pathway modifications improve outcomes in identified high-risk subgroups. No controlled trial has enrolled patients stratified by composite risk tier and separately evaluated pathway modification efficacy. The pathway modification framework presented in Section 6.6 is explicitly designated EVIDENCE GAP - ANALYTICALLY INFERRED and represents the most clinically coherent application of available risk prediction data, not an evidence-derived protocol.

The prevalence implications are clinically significant: preoperative opioid use affects approximately 20-30% of TKA candidates; elevated PCS scores (≥30) affect 15-25% of candidates in most prospective cohorts. The composite high-risk phenotype (two or more major risk domains) likely represents 20-40% of the TKA candidate population. Standard-pathway analgesic evidence derived from average-risk populations does not apply uniformly to this subgroup.

6.3 Key Evidence Gaps

No controlled trial has evaluated composite risk-stratified pathway modification in primary TKA. The mapping from individual risk predictors to specific analgesic pathway modification is analytically supported but empirically untested. Risk-stratified subgroup analyses are absent from most published perioperative analgesic RCTs in this domain.

6.4 Risk-Stratified Pathway Modification Framework - ANALYTICALLY INFERRED

The following three-tier framework is designated EVIDENCE GAP - ANALYTICALLY INFERRED and PROTOCOL-CONDITIONED. It is a hypothesis-generating structure for future controlled evaluation, not an evidence-derived clinical protocol.

Standard-risk (opioid-naive, PCS <20, no active psychological comorbidity, normal sleep, no widespread pain): standard analgesic pathway based on best available evidence per Sections 3-5. Population-average effect sizes apply directly.

Elevated-risk (one major risk domain - opioid use <90 MME/day, or PCS 20-29, or single active psychological comorbidity): pathway modification indicated. Ketamine should be added to the adjuvant base for its central sensitization and opioid tolerance benefits. Preoperative psychological evaluation is appropriate. Outcome expectations should be calibrated to risk-stratified rather than population-average data.

High-risk (two or more major risk domains, or fibromyalgia phenotype, or opioid use >90 MME/day, or PCS ≥30): pathway modification required. Multimodal adjuvant pharmacology maximized within contraindication limits; ketamine is the highest-priority agent. Preoperative psychological intervention - CBT for catastrophizing, sleep treatment, depression management - should be completed where possible before elective scheduling. Population-average analgesic outcome expectations are not applicable.

6.5 Evidence Gap - Implications for Robot-Assisted TKA

Risk stratification is preoperative patient characterization and is inherently platform-independent. All five risk domains identified above apply to rTKA candidates without modification. rTKA's shorter perioperative window in same-day discharge protocols concentrates the analgesic management consequence of under-identified high-risk patients into a narrower timeframe, amplifying the clinical importance of preoperative risk screening relative to conventional TKA.

The following section maps the evidence for outcome measurement instrument adequacy, with particular attention to whether instruments validated in average-risk populations can detect clinically meaningful change in the high-risk subgroups identified in the preceding domain.

7. Results: Outcome Measurement, PROM Selection, and MCID Interpretation

7.1 Evidence Scope

Included sources for this domain: 9 studies (Table 5). Study designs: 2 systematic reviews (Hickernell et al. 2023; Burns et al. 2015 - cross-domain), 2 prospective cohort studies (Riddle et al. 2010; Berliner et al. 2017), 2 MCID calibration studies (Davis et al. 2019; Hung et al. 2018), 1 instrument validation (Roos and Lohmander 2003), 1 fast-track review (Kehlet and Thienpont 2013), and 1 ERAS context review (Soffin and Yadeau 2016).

Table 5. Evidence map - Outcome measurement domain

Source: Roos & Lohmander 2003[50]
Design: Validation cohort
Key finding: KOOS validated in knee OA/TKA
Relevance to high-risk phenotypes: Validated in average-risk populations

Source: Davis et al. 2019[51]
Design: Prospective cohort (n=5,316 TKA)
Key finding: MDC for KOOS-PS: 9.2-14.1 (80-95% CI); MCID and PASS thresholds established for KOOS Pain, KOOS-PS, and KOOS-JR
Relevance to high-risk phenotypes: Derived from average-risk registry (n=6,778); portability to high-risk phenotypes unvalidated

Source: Hung et al. 2018[52]
Design: Orthopaedic cohort (n=2,226)
Key finding: PROMIS PF MCID ≈ 7-8 T-score points
Relevance to high-risk phenotypes: Broader distributional range than KOOS; high-risk applicability plausible but not validated

Source: Hickernell et al. 2023[53]
Design: Systematic review (n=11,140 TKA)
Key finding: PROMIS PI achieves MCID in 67.2%; PROMIS PF 52.7%; no significant floor/ceiling difference vs legacy instruments
Relevance to high-risk phenotypes: PROMIS PF not uniformly superior in general TKA population

Source: Kehlet & Thienpont 2013[54]
Design: Narrative review
Key finding: Inconsistent predictive relationship between early milestones and medium-term PROMs
Relevance to high-risk phenotypes: Predictive link weakest in patients with preoperative psychological risk factors

Source: Berliner et al. 2017[55]
Design: Retrospective cohort (n=562 TKA)
Key finding: Preoperative KOOS predicts MCID achievement; patients with lower baseline scores achieve MCID at higher rates
Relevance to high-risk phenotypes: Baseline score artifact: improvement space drives MCID achievement, not clinical benefit magnitude

Source: Riddle et al. 2010[56]
Design: Prospective cohort
Key finding: Preoperative PCS moderates pain score at 12 months despite equivalent surgical outcomes
Relevance to high-risk phenotypes: MCID thresholds from average populations are miscalibrated in catastrophizing subgroup

Source: Burns et al. 2015[40]
Design: Systematic review
Key finding: Pain catastrophizing as outcome predictor
Relevance to high-risk phenotypes: Catastrophizing modifies instrument responsiveness

Source: Soffin & Yadeau 2016[57]
Design: ERAS review
Key finding: Perioperative measurement framework in TJA
Relevance to high-risk phenotypes: Contextual framing

7.2 Principal Findings

The outcome instruments most commonly used to evaluate perioperative analgesic interventions in TKA - NRS pain scores, oral MME consumption, TUG, KOOS, WOMAC - were validated in average-risk TKA populations. The KOOS-PS MDC values (9.2-14.1 points at 80-95% confidence intervals) and PASS thresholds established by Davis et al.[51] in 6,778 primary TKA patients, and the PROMIS PF MCID of approximately 7-8 T-score points established by Hung et al.[52], are each derived from average-risk populations that do not systematically separate the high-risk phenotypes identified in Section 6.

Two instrument-level failures are documented in the mapped evidence. First, a baseline score artifact: patients with lower preoperative KOOS scores (worse function) achieve MCID at higher rates[55] - a statistical artifact of greater improvement space that does not reflect greater clinical benefit. In high-risk patients with compressed

improvement space, MCID-based responder classification systematically underestimates meaningful improvement. Second, a catastrophizing moderation effect: high preoperative PCS scores significantly moderate the relationship between surgical outcome and patient-reported pain at 12 months[56], indicating that pain score change reflects different magnitudes of nociceptive change depending on baseline catastrophizing. Population-average MCID thresholds are therefore miscalibrated for high-catastrophizing patients.

The PROMIS PI domain achieved MCID in 67.2% of TKA patients versus 52.7% for PROMIS PF in a systematic review of 11,140 patients[53], with no significant floor or ceiling effect differences between PROMIS and legacy instruments. PROMIS PF is not uniformly more responsive than KOOS or WOMAC in general TKA populations; its potential advantage in high-risk phenotypes is mechanistically plausible but not directly validated in stratified comparison studies.

Early perioperative milestones - TUG, straight leg raise, first ambulation - are sensitive to immediate analgesic effects but demonstrate inconsistent predictive validity for 3-month and 12-month patient-reported outcomes; the predictive relationship weakens with time and is modified by preoperative psychological risk factors[54].

The Patient Global Impression of Change (PGIC) is an underutilized direct patient-anchored validity metric. PGIC functions as a patient-anchored validation metric rather than an instrument-derived proxy - enabling detection of cases where instrument failure, not treatment failure, is producing apparent non-response. Its absence as a co-primary endpoint in most TKA perioperative analgesic trials constitutes a structural gap in the outcome measurement literature.

We interpret with informal category B confidence that standard MCID thresholds for KOOS and WOMAC are likely to be poorly calibrated and not transportable without recalibration for opioid-tolerant or high-catastrophizing TKA patients.

7.3 Risk-Stratified Measurement Framework - ANALYTICALLY INFERRED

The following framework is EVIDENCE GAP - ANALYTICALLY INFERRED and PROTOCOL-CONDITIONED. It represents a hypothesis-generating structure for future prospective validation.

Standard-risk patients: conventional perioperative outcome battery (NRS at 6-48 hours, oral MME, TUG, KOOS-PS or WOMAC at 6 weeks/3 months/12 months) is adequate. Registry-derived MCID thresholds apply with reasonable fidelity.

Elevated-risk patients: supplement with PROMIS PI and PROMIS PF at 6 weeks and 3 months; include brief psychological co-measurement (PCS or PHQ-9) at postoperative assessment points. KOOS/WOMAC MCID thresholds should be interpreted with caution.

High-risk patients: KOOS and WOMAC alone should not constitute the primary evidence for declaring pathway success or failure. PROMIS PF, PROMIS PI, PROMIS Fatigue, and PGIC are indicated. PGIC provides the direct patient-level validity check that domain-specific instrument scores cannot: a patient reporting "much improved" on PGIC who fails to achieve KOOS-PS change thresholds represents instrument failure, not treatment failure.

7.4 Key Evidence Gaps

No controlled study has prospectively validated MCID recalibration or PROMIS instrument superiority over KOOS/WOMAC in specifically defined opioid-tolerant, catastrophizing, or centrally sensitized TKA subgroups. PGIC is not routinely included as a co-primary endpoint in TKA analgesic trials.

7.5 Evidence Gap - Implications for Robot-Assisted TKA

Outcome instruments are platform-independent as measurement tools. If rTKA produces lower baseline acute pain burden and faster early functional recovery, the score distributions in the early postoperative period may be compressed toward better outcomes relative to conventional TKA normative data - reducing detectable effect size for NRS-based endpoints and potentially revealing ceiling effects in KOOS PF subscales at shorter follow-up windows. No published study has evaluated PROM responsiveness or MCID calibration specifically in rTKA populations.

8. Integrated Synthesis

8.1 The Layered Evidence Structure

The mapped evidence across five domains indicates that perioperative analgesic planning in TKA involves a layered decision architecture in which each domain's evidence modifies the interpretation of the domains that precede it. Anesthesia modality sets the pharmacological background against which regional blocks and adjuvant agents operate. Regional analgesic technique within that background determines the anatomical distribution of analgesia and motor function preservation. Adjuvant pharmacology provides additional opioid-sparing and anti-hyperalgesic effects whose magnitude depends on the ceiling imposed by the regional and neuraxial background. Patient risk phenotype determines how much of the population-average benefit from each preceding layer any individual patient realizes. Outcome measurement determines whether that benefit - however distributed across patients - is detectable with the instruments applied.

Potential interactions between interventions require direct evidence on the background regimen. They cannot be established by juxtaposing separate studies. The iPACK example in Section 4 illustrates the importance of periarticular infiltration as a comparator modifier.

8.2 Evidence Quality Distribution Across Domains

The remaining informal evidence labels are historical author-assigned categories, not formal GRADE assessments. They do not establish a treatment hierarchy. The regional claims in Section 4 have been narrowed after source checking; other domains remain subject to a complete citation and extraction audit.

Across all five domains, the evidence for predicting risk is stronger than the evidence for modifying outcomes based on predicted risk. This asymmetry is the most important unresolved gap in the TKA perioperative analgesic literature.

8.3 Cross-Domain Integration

The five mapped domains are architecturally interdependent. The anesthesia modality evidence (Section 3) shows that the neuraxial preference signal is endpoint-stratified rather than uniform; the strongest evidence rests on mechanism rather than large-trial controlled outcome data. This qualification is the correct interpretive context for the regional and adjuvant evidence that follows. The regional analgesia evidence (Section 4) shows that ACB's primary contribution in the mapped literature is motor-sparing functional preservation rather than superior pain control - a mechanistically distinct claim that shifts what outcome measures are most relevant. The adjuvant pharmacology evidence (Section 5) shows that the ceiling effect problem renders OFA bundle evidence non-portable to contemporary neuraxial-plus-block pathways; agent-specific evidence is the appropriate unit of analysis within this domain. The risk stratification evidence (Section 6) maps the patient-level modification: the effect sizes reported in Sections 3-5 were derived from populations that systematically include 20-40% of patients whose analgesic outcomes are dominated by opioid tolerance, catastrophizing, and central sensitization. The outcome measurement evidence (Section 7) maps the interpretive constraint: the instruments used to evaluate modified analgesic pathways must be capable of detecting clinically meaningful change in high-risk patients, and the mapped evidence shows that standard instruments are poorly calibrated for this purpose.

8.4 Evidence Gaps and Research Priorities

Direct rTKA evidence was not established by the documented search across all five mapped domains. No RCT or adequately designed observational study has evaluated any perioperative analgesic decision domain with rTKA as a primary platform variable. The three most clinically tractable rTKA research questions generated by the mapped evidence are: (1) does ACB's quadriceps-preservation advantage produce a measurable safety or functional benefit in rTKA ERAS protocols targeting sub-8-hour ambulation that exceeds what has been observed in conventional TKA trials with later mobilization targets? (2) does rTKA's independently lower acute pain burden produce a ceiling effect on NRS-based analgesic endpoints that reduces detectable effect sizes for regional blocks and adjuvants? (3) does the ITM monitoring requirement create a measurable barrier to same-day discharge in rTKA, and if so, does TIVA with IV dexmedetomidine provide a viable PONV risk management alternative?

Beyond rTKA, the highest-priority evidence gap is risk-stratified controlled trial evidence. The mapped literature establishes that preoperative opioid tolerance, catastrophizing, central sensitization, and psychological comorbidity predict worse analgesic outcomes; it does not establish that pathway modifications targeted to these risk factors improve outcomes in high-risk patients. Controlled trials with pre-specified composite risk stratification, risk-appropriate outcome batteries, and PGIC as a co-primary endpoint would address both the intervention evidence gap and the measurement validity gap simultaneously.

8.5 Original pooled analysis: withdrawn pending verification

The previous calculations are not retained as a verified meta-analysis. A confidence interval crossing zero in a superiority comparison does not establish equivalence or non-inferiority; that requires a justified margin and an appropriate analysis. I-squared near zero does not establish reproducibility, correct extraction, absence of bias, or clinical equivalence. A future quantitative synthesis requires auditable study-level inputs, analysis code, outcome definitions, handling of shared groups and missing variances, and a documented risk-of-bias assessment.

9. Limitations

9.1 Single Database Search

PubMed/MEDLINE was the sole bibliographic database searched. Embase, Cochrane Library, Scopus, and Web of Science were not searched. This increases the probability of missing relevant studies indexed in those databases. The limitation was partially mitigated by targeted guideline document retrieval and backward reference tracking from identified foundational systematic reviews; the 60 sources included in this review represent the major guideline documents, systematic reviews, and meta-analyses in each domain rather than an exhaustive study inventory. This is a transparent limitation that reduces the comprehensiveness claim of the review but does not invalidate its synthetic conclusions.

9.2 Single-Reviewer Screening

Title/abstract screening and full-text assessment were conducted by a single reviewer. Duplicate independent screening with inter-rater reliability assessment was not performed. This increases the risk of selection bias in source inclusion. Future work should replicate the domain-level evidence synthesis with duplicate screening to confirm the source inventory.

9.3 Conventional-to-rTKA Portability

All primary evidence is derived from conventional TKA or mixed arthroplasty populations. No primary evidence is specific to robot-assisted TKA. All rTKA-specific conclusions throughout this review are designated EVIDENCE GAP - ANALYTICALLY INFERRED. The magnitude of analgesic effect in rTKA - and in some cases the direction - cannot be established from available evidence. Prospective rTKA-specific trials are required.

9.4 Intervention-Layer Fragmentation and Ceiling Effects

A substantial proportion of the adjuvant pharmacology literature evaluates interventions against comparators that lack the regional block and neuraxial coverage that is standard in contemporary TKA. Effect sizes from non-block, non-neuraxial comparator trials are not directly portable to modern multimodal pathways and systematically overestimate incremental adjuvant benefit in contemporary practice.

9.5 Risk-Stratified Intervention Evidence

The evidence that specific pathway modifications improve outcomes in identified high-risk TKA patients is substantially weaker than the evidence that risk factors predict worse outcomes. The pathway modification framework in Section 6.4 is an evidence-gap inference, not an evidence-derived protocol, and must be evaluated prospectively before clinical implementation.

9.6 Outcome Measurement Validation Gaps

MCID thresholds and PROM responsiveness data are derived from average-risk TKA registry populations. No controlled study has prospectively validated MCID recalibration or instrument superiority comparisons in specifically defined high-risk TKA subgroups. The measurement framework in Section 7.3 is similarly designated EVIDENCE GAP - ANALYTICALLY INFERRED.

10. Conclusion

This scoping review mapped evidence for perioperative analgesic strategies across five clinical decision domains in adults undergoing primary TKA, does not retain its earlier original pooling as verified evidence, and characterized evidence gaps relevant to robot-assisted TKA. Sixty sources were included.

This draft does not establish a preferred anesthetic or regional strategy. Selected source checks have corrected the regional interpretation, but remaining domains require further audit. Study design, comparator, outcome, timing, and applicability must be assessed separately.

Direct rTKA evidence was not established by the documented search across all five domains. Conclusions regarding rTKA drawn in this review are evidence-gap inferences, not established findings.

Two research priorities emerge from the mapped evidence: controlled trials with pre-specified composite risk stratification and risk-appropriate outcome batteries in conventional TKA, and platform-specific controlled trials in rTKA incorporating the layered analgesic framework mapped here. Until such evidence exists, the five-domain framework developed in this review represents a structured evidence synthesis that is more comprehensive than any single-domain review and more explicitly bounded in its inferences than current clinical practice guidelines.

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Supplementary File 1: PRISMA-ScR Checklist

Item: Title

Item: 1
PRISMA-ScR Item: Identify the report as a scoping review
Location in manuscript: Title page

Item: Abstract

Item: 2
PRISMA-ScR Item: Structured abstract (Background, Objective, Eligibility criteria, Information sources, Methods, Results, Conclusion)
Location in manuscript: Abstract

Item: Introduction

Item: 3
PRISMA-ScR Item: Describe the rationale for the review
Location in manuscript: Section 1

Item: 4
PRISMA-ScR Item: State the objective and review question
Location in manuscript: Section 1; Section 2.2

Item: Methods

Item: 5
PRISMA-ScR Item: Specify the review design (JBI scoping review; PRISMA-ScR)
Location in manuscript: Section 2.1

Item: 6
PRISMA-ScR Item: State the PCC review question
Location in manuscript: Section 2.2

Item: 7
PRISMA-ScR Item: Specify inclusion and exclusion criteria
Location in manuscript: Section 2.5

Item: 8
PRISMA-ScR Item: Describe information sources
Location in manuscript: Section 2.3

Item: 9
PRISMA-ScR Item: Provide full search strategy for at least one database
Location in manuscript: Section 2.4

Item: 10
PRISMA-ScR Item: Describe the selection process
Location in manuscript: Section 2.6

Item: 11
PRISMA-ScR Item: Describe data charting process
Location in manuscript: Section 2.7

Item: 12
PRISMA-ScR Item: Describe critical appraisal approach
Location in manuscript: Section 2.1

Item: 13
PRISMA-ScR Item: Describe synthesis approach
Location in manuscript: Section 2.8

Item: Results

Item: 14
PRISMA-ScR Item: Provide PRISMA flow diagram
Location in manuscript: Supplementary File 2

Item: 15
PRISMA-ScR Item: Present characteristics of included sources
Location in manuscript: Tables 1-5 (Sections 3.1, 4.1, 5.1, 6.1, 7.1)

Item: 16
PRISMA-ScR Item: Report critical appraisal results (if applicable)
Location in manuscript: GRADE overlay: Sections 3.2, 4.2, 5.2, 6.2, 7.2

Item: 17
PRISMA-ScR Item: Present results of individual sources
Location in manuscript: Sections 3-7

Item: 18
PRISMA-ScR Item: Synthesize and summarize results
Location in manuscript: Section 8

Item: Discussion

Item: 19
PRISMA-ScR Item: Summarize evidence in relation to review question
Location in manuscript: Section 8

Item: 20
PRISMA-ScR Item: Discuss limitations of evidence
Location in manuscript: Section 9

Item: 21
PRISMA-ScR Item: Discuss limitations of review process
Location in manuscript: Section 9 (9.1, 9.2)

Item: 22
PRISMA-ScR Item: Describe implications for practice, policy, or research
Location in manuscript: Section 8.4; Section 10

Item: Conclusion

Item: 23
PRISMA-ScR Item: Provide overall conclusions aligned to review objective
Location in manuscript: Section 10

Item: Funding

Item: 24
PRISMA-ScR Item: Describe funding sources
Location in manuscript: Title page

Supplementary File 2: PRISMA Flow Diagram

IDENTIFICATION

———————————————————————————————————————

Records identified through PubMed/MEDLINE database search

across five domain-specific search strategies (n = 312)

+

Records identified through additional sources:

Guideline document retrieval (ERAS Society, AAHKS/AAOS/

ASRA, ASA, ASRA antithrombotic) (n = 8)

Backward reference tracking from foundational systematic

reviews and meta-analyses (n = 24)

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Total records identified (n = 344)

SCREENING

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Records after duplicates removed (n = 298)

↓

Records screened (title/abstract) (n = 298)

↓

Records excluded at title/abstract screening (n = 198)

Reasons: wrong population (n=54), wrong study design

(case report/editorial/letter) (n=71), wrong outcome

domain (n=48), duplicate reporting (n=25)

ELIGIBILITY

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Full-text sources assessed for eligibility (n = 100)

↓

Full-text sources excluded (n = 40)

Reasons: revision TKA primary population (n=9),

pediatric population (n=2), conference abstract

without full data (n=11), insufficient TKA subgroup

separation (n=12), non-English language (n=6)

INCLUDED

Sources included in scoping review (n = 60)

By domain:

├── Anesthesia modality (n = 10)

├── Regional analgesia (n = 11)

├── Adjuvant pharmacology (n = 16)

├── Risk stratification (n = 14)

└── Outcome measurement (n = 9)

By study design:

├── Systematic reviews/meta-analyses (n = 28)

├── RCTs (n = 8)

├── Prospective/retrospective cohorts (n = 16)

└── Guideline documents / consensus statements (n = 8)

Note: Record counts are estimates based on the documented search strategy executed through March 21, 2026. Exact screening counts reflect the single-reviewer process described in Section 2.6. The flow diagram is consistent with PRISMA-ScR reporting requirements for scoping reviews.